

ScalpShield – Production AI Risk Classification System

Project Summary

ScalpShield is a production-style AI system designed to automate fraud and risk detection in ticketing workflows. The system ingests structured data, applies a trained machine learning model, and exposes real-time predictions through a REST API consumed by a web dashboard, enabling actionable decision-making within business processes.

System Architecture & Workflow

- FastAPI backend providing a REST endpoint for automated risk scoring
- CSV-based data ingestion with schema validation and structured error handling
- XGBoost model loaded at runtime for low-latency inference
- JSON responses designed for integration into downstream applications
- Next.js dashboard for visualization and review of AI outputs

AI & Data Engineering

- Built a feature engineering pipeline using Pandas and NumPy to capture abnormal behavioral patterns
- Trained and evaluated a supervised XGBoost classifier on synthetic fraud scenarios
- Implemented confidence-based risk tiers (green, yellow, red) to support automated workflows
- Performed validation and error analysis to assess model reliability

Engineering Ownership

- Led backend and ML development on a 6-person team
- Designed the full inference pipeline from data ingestion to API response
- Implemented request validation using Pydantic and file handling via python-multipart
- Integrated backend services with a TypeScript-based Next.js frontend

Tech Stack: Python, FastAPI, Uvicorn, XGBoost, Pandas, NumPy, Pydantic, Next.js, React, TypeScript, Tailwind CSS, Recharts